Recall, not verification, is the bottleneck when frontier LLMs elucidate molecular structures from real spectra

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Abstract

Given a real, unedited infrared and NMR spectrum from the published literature, how often does a frontier large language model (here, Claude) recover the correct molecular structure? We find: **28% of the time** (top-1, n=194; 95% CI 22–35) — far below the near-100% implied by curated demonstrations. The bottleneck is not the model's judgment but its *proposal*: when the true structure is among the candidates, forward-verification selects it **84%** of the time, yet the model generates it for only **31%** of compounds. **Recall, not verification, is the wall.** We establish this with three contributions. First, **IRexp** — the largest permissively-licensed, redistributable dataset of *experimental* infrared spectra (121,233 records; 43,060 structure-linked; 33,201 full IR + ^{1}H + ^{13}C + structure quadruples), mined from open-access literature. Second, **IRSpectra-Bench** — an open, blind, mechanically scored benchmark of 194 compounds, complexity-stratified, on which accuracy splits sharply (48% simple vs 8% complex); a within-compound control shows that solving each compound in a fresh context with tools increases accuracy over one long context (5%→15%, i.e. 1/20→3/20; directional at n=20), so reported numbers are sensitive to methodology, not raw capability alone. Third, a training-free **forward-verification** method that acts on the diagnosis: on the 60-compound forward-verify subset, generating regiochemistry-aware candidates and re-ranking by predicted-vs-observed ^{13}C lifts top-1 from 23% to **30%** — a real but bounded gain: within the training-free regime, accuracy is capped by the product of generation recall and verification precision. That ceiling is *elicitation-specific, not intrinsic*: a small generator fine-tuned on IRexp lifts recall from 41% to 56% (SI; with a corresponding rise in forward-verified top-1, pending deposit of the forward-prediction step), and the released data is the active ingredient (a simulated-pretrained model recovers 0/248 structures zero-shot, 25% after IRexp fine-tuning). The recall-bound regime reproduces in a battery-electrolyte domain (n=46, 26% top-1). Every result is single-vendor (Claude); a cross-vendor test is the key open question. The core protocol uses no model training and no paid API — two clearly-fenced probes (a trained generator, §5.6, and a learned verifier, §5.7) are the only exceptions — and all data, predictions, and code are released.

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1. Introduction

Determining a molecule's structure from its spectra is a central, time-consuming task in synthetic and analytical chemistry. The dominant machine-learning approach trains specialised encoder–decoder models to map spectra to structures; the recent *Spectro* model, for example, learns ^{1}H/^{13}C/IR → SELFIES from a corpus of 6,833 molecules.[1] In parallel, general-purpose LLMs have been reported to perform the same task off-the-shelf: a 2026 industrial white paper (a non-peer-reviewed company report) found that Claude Opus matched or beat commercial NMR-prediction software in the forward direction (structure→spectrum, ±0.08 ppm ^{1}H) and "recovered all eight simpler structures on every attempt" in the inverse direction (spectrum→structure).[2] We treat its numbers as a motivating claim to be tested against peer-reviewed benchmarks, not as an established baseline.

These results are striking, but the evaluation that supports them is narrow: 15 inverse problems on curated single-ring or two-fragment molecules, NMR only, with the seven harder targets additionally given the *starting-material structure* as a hint, and "recovery" scored leniently over three runs and three ranked candidates. The question a practising chemist actually faces — *take an arbitrary experimental spectrum from a paper and recover the structure* — is left open. Answering it requires (i) a large, diverse, real benchmark; (ii) a blind, reproducible scoring protocol; and (iii) honest accounting for the methodological choices that inflate or deflate the apparent score.

We provide all three. We build IRexp (§2), a literature-mined dataset that supplies the IR modality absent from prior LLM evaluations alongside paired NMR and resolved structures. We define a blind benchmark on it (§3) and measure inverse-elucidation performance under matched and mismatched methodologies (§4), isolating a large solver-methodology effect with a within-compound control. We then introduce forward-verification elucidation (§5), a training-free method that turns the model's *strong* direction (forward prediction) against its *weak* one (inverse regiochemistry), and use it to localise the bottleneck. The central finding is a sharp asymmetry: given a candidate set that contains it, the model *verifies* the correct structure 84% of the time, yet *proposes* it for only 31% of compounds — recall, not verification, is the wall (Fig. 1). The end-to-end study design is summarised in Fig. S1. Throughout, the solver and verifier are LLM agents run under a consumer subscription — no fine-tuning, no API spend — making the core protocol cheaply reproducible; two trained probes (§5.6, §5.7) are reported separately as fenced complements. By construction this is an open-resource contribution in the remit of *Digital Discovery*: an openly licensed, structure-linked spectral dataset; a pre-registered, mechanically scored benchmark with released ground truth and scorer; and a training-free, zero-paid-API core pipeline (with two trained probes fenced in the SI) in which every figure regenerates from the released code and predictions. We prioritise an honest, reusable measurement of where current models stand over a leaderboard-topping number.

Contributions. 1. **IRexp** — the largest openly *redistributable* experimental-IR dataset (121,233 records; 43,060 structure-linked; 33,201 IR+^{1}H+^{13}C+structure): larger by raw record count than any prior collection and uniquely released under permissive licences for bulk model development (view-only resources such as SDBS are larger in structure-linked spectra but are not redistributable; §2.1), with a reproducible mining pipeline. 2. An **open multimodal benchmark** and the first blind, mechanically scored, complexity-stratified evaluation of frontier-LLM structure elucidation on real data — measured in depth on one model family (Claude) — reconciling the gap to optimistic prior reports. Crucially, the ground truth is experimental structures from the published literature, resolved deterministically (OPSIN/RDKit) and expert-audited: no LLM curates the labels or scores the predictions. The model is the system under test, not the source of its own answers. 3. A **diagnostic decomposition** of LLM elucidation into recall and verification, showing that **recall — not verification — bounds current performance**, obtained via a training-free forward-verification probe that yields a bounded improvement (top-1 23%→30%) but, by our own measurements, cannot exceed a recall/precision ceiling without sharper verification or 2D-NMR data.

1.1 Related work

Trained spectra→structure models. The dominant line trains sequence or graph decoders to emit structures from spectra: *Spectro* (^{1}H/^{13}C/IR→SELFIES)[1], the multitask CNN+transformer of routine 1D-NMR[3], and set/graph transformers such as NMRTrans[4]. These reach high accuracy *in-distribution* but require a labelled spectra→structure corpus, which is exactly the scarce resource our IRexp pipeline targets, and they are retrained per modality. Closest in spirit to our multimodal setting is **NMIRacle**[5], a generative model conditioned jointly on IR + ^{1}H + ^{13}C; it is a strong trained baseline, whereas our contribution is the *open experimental-IR data it (and others) can train on*, plus a *training-free* protocol and a blind benchmark to measure it.

LLMs as elucidators, and how this work differs. General-purpose LLMs have been applied off-the-shelf — Anthropic's forward/inverse white paper[2], SpectraLLM and MolSpectLLM (multimodal LLMs over multi-spectral input)[6,7], and knowledge-enhanced tree-search reasoning[8] — and dedicated multimodal *benchmarks* now exist, most prominently **MolPuzzle**[9] (IR+MS+^{1}H+^{13}C elucidation puzzles with the molecular formula supplied). These establish that the task is worth benchmarking and that LLMs can be scored on multi-spectral input; we claim priority on neither. What is new here is three things that, to our knowledge, no prior benchmark combines. **(i) Real, literature-mined experimental spectra at scale:** prior multimodal benchmarks and trained baselines[1,5,10] rely on simulated, software-predicted, or hand-curated puzzle spectra, whereas IRSpectra-Bench is drawn from IRexp's experimental IR + ^{1}H + ^{13}C of out-of-distribution compounds, so it measures the simulated-to-real gap rather than in-distribution skill. The closest trained precedent, Alberts et al.[10], learns an IR→structure transformer from ~635k *simulated* spectra with a 3,453-spectrum experimental fine-tune (top-1 44% on 6–13 heavy atoms); IRexp instead releases experimental IR at scale as an open, redistributable resource and pairs it with a blind LLM benchmark. **(ii) Blind, fully specified, mechanical scoring:** reported accuracies on the same task vary enormously with inference method and scoring harness — GPT-4o on MolPuzzle ranges from 1.4%[9] to 27.8%[8], the latter using knowledge-enhanced tree-search reasoning[8] — so numbers across papers are not directly comparable; we therefore fix and release a single, pre-registered, RDKit-InChIKey protocol with bootstrap CIs. **(iii) A recall/verification decomposition:** existing benchmarks report a single aggregate score, whereas we factor performance into generation recall and verification precision and show *where* the task is lost — the result that transfers across models, domains, and verifier choices.

Computational NMR for structure validation. Our forward-verification method is the LLM analog of a workflow chemists already trust: assign a structure by computing the spectrum each candidate *would* give and matching it to experiment. In solution, this is the DP4 / DP4+ probabilistic framework over GIAO-DFT shifts[11,12]; in the solid state it is **NMR crystallography**, where GIPAW-computed shifts adjudicate between candidate structures[13,14]. We replace the quantum-chemical predictor with a forward LLM, trading accuracy for zero setup cost, and inherit the same core principle — *verification by forward prediction is easier than inverse generation.*

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2. The IRexp dataset

2.1 Motivation

Open experimental IR is scarce *in a redistributable, ML-ready form.* The largest freely *downloadable* collections are the NIST WebBook[15] (~16k IR spectra) and the Chemotion electronic-lab-notebook[16] deposit (~2k). The AIST SDBS[17] is larger (~54k FT-IR, all structure-linked) but is *view-only*: it caps downloads at 50 spectra/day, forbids commercial use, and ships no permissive licence or bulk export, so it cannot be used to train or redistribute open models at scale. Commercial libraries (e.g. Wiley KnowItAll, ~10^{5} spectra) are larger still but closed. IR — which directly encodes functional groups complementary to NMR — has thus had no large, *permissively-licensed*, structure-linked, bulk-reusable resource. IRexp fills this gap, and to our knowledge is the largest *openly redistributable* experimental-IR collection by raw record count.

2.2 Construction

A per-compound IR band list, together with co-reported ^{1}H/^{13}C NMR, follows a remarkably stable textual convention in the experimental sections of organic chemistry papers. We exploit this with a browser-free harvesting agent:

- **Discovery.** Open-access primary literature is enumerated through the NCBI E-utilities and harvested in bulk from the PMC Open-Access Subset[18] on AWS S3 (plain HTTPS, no anti-bot, fully redistributable CC-BY content), supplemented by the Chemotion FT-IR deposit (RADAR4Chem, CC-BY-SA-4.0).
- **Extraction.** A deterministic parser segments experimental text into per-compound records and extracts IR wavenumbers and ^{1}H/^{13}C shift lists, with quality gates that reject instrument scan-range artefacts and prose false-positives (band-list density, ≥4 bands, plausible 400–4000 cm^{-1} window).
- **Structure resolution.** In-text IUPAC names are converted to SMILES with OPSIN[19], canonicalised with RDKit[20] (InChIKey, SELFIES[21]), with a PubChem[22] fallback for trivial/natural-product names. A key engineering finding was that the dominant open-access main-text convention labels compounds with *letter-prefixed* series labels (e.g. "…carbothioamide (**B1**):") rather than the digit-first "(3a)" of supporting-information sections; capturing these and cleaning narrative/PDF artefacts before resolution raised structure coverage from 24% to **35%**.

The pipeline is browser-free by design. This is not an anti-bot bypass — the bulk corpus is fully open — but a throughput choice: plain HTTP from a bulk corpus parses at hundreds of papers per second, which is what makes a 10^5-scale harvest feasible where a per-page browser driver (as used to build prior 10^3-scale sets) is not.

2.3 Contents and licensing

Table 1. IRexp dataset contents and provenance.

field	value
experimental IR records	**121,233**
…co-reporting ^{1}H and/or ^{13}C NMR	87,075 (72%)
…with a resolved 2D structure	**43,060 (35.5%)**
…of these, with ^{1}H and/or ^{13}C NMR	**40,702**
…of these, with both ^{1}H and ^{13}C (full quadruples)	**33,201**

A structure-complete split, `irexp_resolved` (43,060 records, 100% structure-linked), is the training-/benchmark-ready subset and is ~6× the 6,833-molecule set used to train Spectro[1] (Fig. S2). Provenance is 119,345 PMC-OA records (CC-BY) plus 1,888 Chemotion records (CC-BY-SA); the two licences are kept as separable pools. Each record is DOI-/accession-traceable. Re-resolution is additive and content-keyed, so the dataset can be re-enriched without re-mining.

3. Benchmark design (IRSpectra-Bench)

From `irexp_resolved` we draw **IRSpectra-Bench**, 194 blind elucidation problems. Each problem presents the **molecular formula** (as from HRMS), the **IR band list**, and the **^{1}H and ^{13}C shift lists** (with multiplicities and J-couplings where reported), and asks for the structure. No name, SMILES, or hint is given. Every main-round ground-truth structure is **spectrally validated** by an automated RDKit consistency check (^{13}C peak count vs symmetry-unique carbons, molecular-formula match, SELFIES round-trip), excluding records with merged or incomplete spectra (6/140 in the main round, leaving 134). The 60 compounds of the controlled rounds are retained as fixed, pre-registered sets for the difficulty and within-compound controls, with the same self-consistency audit reported separately rather than used to exclude (57/60 pass; §7). Problems are stratified by RDKit ring analysis: a compound is **simple** iff it has at most two rings (single ring or two separate ring fragments), no fused/spiro/bridgehead system, and $\leq$22 heavy atoms; every other compound — any fused/spiro/bridged or $\geq$3-ring system, **or** more than 22 heavy atoms — is **complex** (98 simple / 96 complex). This binary difficulty axis is distinct from the continuous size gradient in §4.1, which bins by heavy-atom count ($\leq$15 / 16–25 / >25). The criterion is therefore exhaustive (the 23–24-atom band, 13 compounds, is classed complex by size). InChIKey de-duplication is applied across all rounds to prevent leakage.

Scoring is mechanical. A prediction is *correct* if its RDKit InChIKey connectivity layer (first 14 characters) matches the reference — i.e. we score *constitution* (atom-and-bond connectivity); stereochemistry is reported separately. This is a deliberate floor: a candidate with the correct constitution but wrong stereochemistry counts as correct. The exposure is bounded — only 10.3% (20/194) of benchmark answers carry a *defined* (assigned R/S) stereocentre (29.9% any potential) — and 1D ^{1}H/^{13}C/IR rarely fixes absolute configuration, so a stereochemistry-aware score is neither well-posed from the given data nor materially different on this set.

Metrics. We report five quantities. **Top-1 (exact constitution)** is the fraction of compounds whose single best-ranked candidate matches the reference at the InChIKey connectivity layer. **Recovered (top-3)** is the fraction for which the reference appears among the up-to-three ranked candidates returned (matching the lenient "recovery" protocol of ref.[2]). **Generation recall** is the fraction for which the reference is present in the candidate pool *before* re-ranking — the ceiling any verifier can reach. **Conditional-on-recall precision** is the verifier's hit rate over recall-positive compounds only, isolating verification quality from generation. The forward verifier ranks candidates by a symmetric **chamfer distance** between predicted and observed ^{13}C peak sets (for each predicted peak, the distance to its nearest observed peak, summed, and symmetrically for each observed peak; lower is better, with no equal-count requirement). We also report Morgan(2, 2048)[23] Tanimoto as a graded "right scaffold/family" signal. CIs are bootstrap 95% over compounds; model-vs-model differences use McNemar's exact test with Holm correction.

Solvers are LLM agents run under a consumer subscription. A frontier LLM (Claude Opus) is invoked as an independent sub-agent per batch of problems; agents are closed-book — configured with no web/tool access beyond an RDKit formula check and no access to ground truth, verified by grep-auditing their task transcripts at run time (audit logs available on request; the committed artefacts are the parsed per-compound predictions). Adherence to the supplied formula is high but imperfect: 91.3% (115/126) of forward-verification candidates match the given molecular formula exactly (100% of true structures, 89.6% of decoys), and 76.6% across the full top-3 pool — a residual generator error orthogonal to regiochemistry. This makes the benchmark free to run and reproducible without API credits.

4. How well do LLMs elucidate real structures?

4.1 Headline performance

Decoupled per-compound agents solve each problem in an independent context (formula + IR + ^{1}H + ^{13}C, blind, up to three ranked candidates). Over the full **194-compound benchmark** (134 spectrally-validated compounds + the 60 from the controlled rounds), with bootstrap 95% confidence intervals:

Table 2. Headline elucidation performance on IRSpectra-Bench (n=194), overall and by difficulty stratum, with bootstrap 95% CIs.

metric	overall (n=194)	simple (n=98)	complex (n=96)
top-1 exact constitution	**28.4%** [22–35]	48.0% [39–57]	8.3% [3–15]
recovered (within top-3)	33.5% [27–40]	54.1% [44–63]	12.5% [6–20]
scaffold-level (best Tanimoto $\geq$ 0.45)	56%	73%	39%
mean best Tanimoto	0.59	0.73	0.45

As a cohort-robustness check, restricting the controlled rounds to their spectrally-clean subsets as well (134 main-clean + 57 controlled-clean = 191 strictly-validated compounds) leaves the headline unchanged: top-1 **28.8%** (95% CI 23–36), recall 34.0%, with simple 49.0% / complex 8.4% — within ~1 point of the n=194 figures on every metric shown, with fully overlapping intervals, so the asymmetric inclusion of the pre-registered controlled sets does not drive the result.

The gradient is sharp and the intervals are tight. The 48%→8% simple→complex separation (Fig. 2) confirms the benchmark is discriminating across a realistic difficulty range, and accuracy falls monotonically with molecular size in step with it — top-1 **60.5%** for ≤15 heavy atoms, 28.3% for 16–25, and **7.0%** above 25 (Fig. S3). The model recovers molecular formula and functional groups reliably and the scaffold often (best Tanimoto $\geq$ 0.45 for 56% of compounds), but the exact constitution far less often, failing predominantly on **regiochemistry** — *which* position a substituent occupies. This is consistent with an information limit of 1D data: many regioisomers have similar ^{1}H/^{13}C shifts, which is precisely why 2D experiments (HMBC, NOESY) exist.

4.2 Reconciling with prior reports

Our 28% top-1 sits far below the ~100% on "simple" molecules reported, in a non-peer-reviewed company white paper, for the same model class.[2] Because that figure has not been independently scored, we reconcile it against the *peer-reviewed* record (the MolPuzzle benchmark[9] and its re-scorings[8], and the trained baselines[1,5]) rather than treating it as settled. The gap is largely attributable to methodology and scoring choices, not raw capability:

- **Difficulty.** "Single ring" by ring-count is not "easy": our simple stratum includes, e.g., a hexasubstituted benzene whose regiochemistry has many realisations. Ring count is a poor proxy for elucidation difficulty.

- **Scoring.** Prior work counts a recovery if the reference appears among three ranked candidates over three independent runs; we report single-run top-1 and top-3.
- **Hints.** Prior hard targets received the starting-material SMILES, which fixes most of the scaffold; we give none.
- **Curation.** Prior compounds were hand-selected; ours are scraped and unfiltered for solvability.

On a like-for-like easy/hinted/lenient setup the numbers rise; on the realistic, hint-free, scraped regime, ~28% is the honest figure.

Versus trained models — a bound, not a leaderboard. A like-for-like comparison against specialised trained models is not available: no system has been scored on an identical test set, and published numbers differ in the three respects that most move the score — spectrum realism (simulated/curated vs. real), hints, and how "exact match" is defined. The strongest trained baselines report their accuracy *in-distribution on simulated spectra*. Spectro ($^1H/^{13}C$/IR→SELFIES, 6,833 training molecules) reaches ~90% top-1 exact recovery — but on a 1,366-molecule held-out split whose IR is plotted from reference data and whose NMR is software-*predicted*, not experimental[1]; and NMIRacle, which like us conditions jointly on IR+^1H+^{13}C with *no* hints, reports 48% top-1 / 66% top-15 exact-SMILES recovery — again on held-out molecules from a *simulated* corpus drawn from the training distribution[5]. We measure 28.4% top-1 (33.5% top-3) on the full n=194 benchmark — rising to 30% with forward verification on the n=60 forward-verify set (§5) — on **blind, real, literature-mined experimental** spectra of out-of-distribution compounds. These are not comparable as a leaderboard; read as a *bound on the simulated-to-real gap*, the contrast suggests that high in-distribution accuracies substantially overstate real-world performance — the same gap we document for the LLM above. The instability of the metric itself reinforces the caution: the same ~20× MolPuzzle swing documented in §1.1 (GPT-4o: 1.4%[9]–27.8%[8], method- and harness-dependent) bounds how much weight any single unaudited near-100% claim[2] can bear.

4.3 Methodology dominates: a within-compound control

The same 20 molecules were solved two ways: (a) by a single LLM context handling all of them sequentially with no tools, and (b) by independent per-compound agents with RDKit formula-checking. On the *identical* compounds, recovery rose from **5% to 15%** (1/20 → 3/20) and top-1 from 0% to 15% (0/20 → 3/20) — a 3× methodology effect with zero sample confound, though on a small set (the 15% point estimate carries a Wilson 95% CI of roughly 5–36%), so it is read as a directional within-compound demonstration rather than a precise multiplier. Small rounds also swing widely (15–40% across n=20–40 draws), which is why the headline is the full **194-compound** figure (28.4% top-1, 95% CI 22–35) rather than any single round. The practical lesson — fresh per-compound context plus tool access roughly triples apparent performance — also explains part of the gap to optimistic prior reports, whose per-problem API calls implicitly used method (b).

4.4 Model comparison: the benchmark ranks capability (and separates the extremes)

A benchmark is only useful if it separates models. On a fixed 24-compound subset solved blind by four Claude models — spanning a wide capability range, including the newest (Fable 5) — under the identical protocol (Fig. 3):

Table 3. Four-model comparison on a fixed 24-compound subset under the identical blind protocol.

model	top-1	recovered	top-1 95% CI
Claude Fable 5	**45%**	54%	[25, 67]
Claude Opus	25%	29%	[8, 42]
Claude Sonnet	20%	25%	[8, 38]
Claude Haiku	**0%**	4%	[0, 14]

CIs are bootstrap except Haiku's, which is the Clopper–Pearson exact interval for 0/24 (the percentile bootstrap is degenerate at a boundary count of zero).

Three signals matter here. First, the four models rank in **monotonic capability order**, and the outcomes are **strictly nested** (Haiku ⊂ Sonnet ⊂ Opus ⊂ Fable — each stronger model solves a superset of the compounds the weaker one does), with the smallest **floored at 0% exact** (4% recovered) on the same problems — so the benchmark is **capability-sensitive** and not a coin-flip. The nesting makes the *ranking* robust, but at n=24 the subset is **underpowered to separate adjacent models**: by McNemar's exact test only the Fable-vs-Haiku gap survives multiple-comparison correction (Holm-adjusted p=0.006); the Fable-vs-Opus gap is not significant (uncorrected p=0.063, Holm-adjusted p=0.19), and Opus vs Sonnet are statistically indistinguishable — the bootstrap CIs above overlap accordingly. Second, two mid-tier frontier models agree closely (Opus 25%, Sonnet 20%), so the recall-bound regime of §4.1 (top-1 in the high-20s%) is **not an artefact of a single model**. Third, the newest model nearly **doubles** the next-best top-1 (45% vs 25% on identical compounds; a large but, at this n, not-yet-significant gap) yet still misses the majority — IRSpectra-Bench is **hard and far from saturated even for the strongest model available**, with clear headroom. We ran four Claude-family models because they are callable for free under one subscription; a true cross-vendor sweep (GPT-, Gemini-class, open models) needs API access we deliberately did without, but the monotonic, large capability spread makes it unlikely the recall-bound pattern is specific to one model lineage. This is the behaviour a benchmark needs to remain informative as models improve.

4.5 Domain case study: battery-electrolyte chemistry

Structure elucidation from spectra is the daily inverse problem of the electrolyte-and-interphase community: the soluble decomposition products of carbonate and sulfone electrolytes, the additives that build the solid-electrolyte interphase, and the fluorophosphate/fluorosulfonyl species that NMR is routinely used to assign are exactly the molecules whose constitution must be read back out of IR and multinuclear NMR. To test whether the recall-bound regime above holds for *this* chemistry — rather than for organic molecules at large — we curated **IRSpectra-Bench-Electrolyte**, a 48-compound subset (eight per class as curated; two later yielded no parseable candidate, leaving **46 scored**) of structure-resolved IRexp records selected by substructure for the six functional families that dominate lithium- and sodium-battery electrolytes and their breakdown products: **carbonate** (linear/cyclic, the EC/DMC backbone), **sulfonyl/sulfonate** (sulfones, sulfonamides, the $-SO_2CF_3$ motif of imide salts), **nitrile** (acetonitrile- and adiponitrile-type high-voltage additives), **sp^3-C–F** (fluorinated solvents and additives), **phosphoryl** (phosphates/phosphonates, the $LiPF_6$-derived OPF chemistry), and **glyme/oligoether** (the ether solvents and PEG linkers). These are literature compounds bearing the *functional chemistry* of electrolytes, drawn from the open corpus; they are **not** operando or in-cell degradation spectra, and we make no claim that they are. They are excluded from every other split in this paper. Each compound was solved blind under the identical decoupled-agent protocol (Opus, closed-book, up to three ranked candidates, RDKit only for formula/parse checks).

Performance lands in the **same broad regime as the headline benchmark** — overall **top-1 26%, recovered 28%** (12/46 and 13/46; two compounds received no parseable candidate; at n=46 the 95% CI is wide and overlaps the headline) — consistent with the bottleneck being a property of the elucidation task rather than of any one chemical neighbourhood. The per-class breakdown (Fig. S4) is itself informative:

Table 4. Per-class performance on IRSpectra-Bench-Electrolyte (n=46).

electrolyte class	n	top-1	recovered (top-3)
sp^3-C–F	8	**50%**	50%
carbonate	7	29%	43%
phosphoryl	8	25%	25%
glyme / oligoether	7	29%	29%
sulfonyl / sulfonate	8	**12%**	12%
nitrile	8	**12%**	12%

At n=7–8 per class these per-class differences are within sampling noise (six-class χ^2 p≈0.56; best-vs-worst Fisher exact p≈0.28), so the ordering is a hypothesis-generating pattern rather than an established ranking. Read that way, it is chemically legible. **sp^3-C–F** centres are the easiest: a C–F coupling fingerprint (the large $^1J/^2J(C,F)$ splittings) localises the fluorine and pins regiochemistry, removing exactly the degeneracy that defeats elucidation elsewhere. **Sulfonyl** and **nitrile** are the hardest, for two distinct reasons that recur in real electrolyte analysis: sulfur/phosphorus **oxidation-state ambiguity** — sulfide vs. sulfoxide vs. sulfone, $-SCF_3$ vs. $-SO_2CF_3$, the very distinctions an interphase study must make — is poorly constrained by $^1H/^{13}C$ shifts alone (it lives in the IR and in heteronuclei the benchmark does not score); and nitrile-bearing targets in the corpus sit on heavily substituted heteroaromatic cores whose regiochemistry the 1D spectra underdetermine. This is the recall-bound failure of §4.1 reappearing in a domain-specific guise, and it is precisely where the **forward-verification** recipe of §5 — compute the spectrum each candidate *would* give and match it to the one observed — plays a role *analogous* to (not a replacement for) computational-NMR / NMR-crystallography structure validation, sharing its forward-prediction logic while trading DFT's calibrated accuracy for zero setup cost. The subset, its per-class answers, and the scorer are released with the benchmark.

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5. Forward-verification elucidation

5.1 Method

The inverse direction is the model's hard, isomer-blind direction; the **forward** direction (structure→spectrum) is its easy, accurate one.[2] Regioisomers, crucially, have *different* forward-predicted ^{13}C shifts. We therefore close a generator– verifier loop:

> **generate** candidate structures (inverse) → **forward-predict** each candidate's ^{13}C spectrum, blind to the observed spectrum → **re-rank** candidates by the distance between predicted and observed ^{13}C → return the best match.

This mirrors how a chemist confirms a structure ("if it were that isomer, C-3 would be at ~120 ppm; we observe 135, so it is the other"), exploits the standard principle that verification is easier than generation, and requires no training. **Fig. 4** shows the mechanism on a real benchmark pair — the picolinamide / nicotinamide regioisomers, which the inverse task cannot separate but whose forward-predicted ^{13}C spectra match the observed one at 0.42 vs 1.30 ppm. We implemented the verifier as independent LLM agents that predict ^{13}C shift lists from SMILES alone, using pure reasoning with no tools; candidates were shuffled and anonymised so that isomers of one target never co-occur. Predicted and observed ^{13}C peak sets were then compared with a symmetric chamfer distance.

For balance — and because conditional precision is 72–84%, not 100% — the verifier also produces clear **false positives** on near-degenerate pairs. On the 2-(nitrophenyl)-2,3-dihydroquinazolin-4(1*H*)-one targets ($C_{14}H_{11}N_3O_3$), the forward predictor cannot separate the *ortho*- and *meta*-nitrophenyl regioisomers: their predicted ^{13}C lie within its ~2 ppm error, and it ranks the wrong (2-nitrophenyl) isomer first by a chamfer of 1.35 ppm versus 1.36 ppm for the true (3-nitrophenyl) one — a 0.01 ppm margin, i.e. effectively a coin flip. This is the precision-loss mechanism of §5.3 made concrete: when candidates are near-degenerate the cheap predictor cannot resolve them, and it is exactly these cases — not generation failures — that a sharper (DFT-level) or 2D-NMR-grounded verifier would need to fix.

Conceptually this is **analogous to NMR-crystallography logic, with an LLM in place of the quantum chemistry** — an analogy, not an equivalence: where DP4/DP4+ rank candidates by GIAO-DFT shifts[11,12] and NMR crystallography adjudicates polymorphs and connectivity by GIPAW-computed shifts[13,14], we rank by the shifts a forward LLM predicts. The shared principle is *verification by forward prediction*; what we do **not** inherit is DFT's calibrated error model. The trade is deliberate — we forgo the ≈1–2 ppm accuracy and the rigorous, probabilistic error model of DFT (the very thing that lets DP4 emit a posterior probability) for a predictor that runs in seconds at zero setup, and we quantify below exactly how far that cheaper, uncalibrated verifier carries the inverse problem.

5.2 Result

On the 60 benchmark compounds (126 candidate structures from the solver agents):

Table 5. Forward-verification decomposition on the 60 forward-verify compounds.

	value
generation recall (true structure among candidates)	**31%**
top-1, solver self-ranking	23%
top-1, **forward-verified re-ranking**	26%
conditional on recall — self-ranking	14/19 (**73%**)
conditional on recall — forward-verification	16/19 (**84%**)

The decomposition is the finding. **When the true structure is among the candidates, forward verification selects it in 16 of 19 cases (84%)** — high in absolute terms and ~18 points above the 66% derangement chance floor of §5.5 (given how few and near-degenerate the recall-positive candidate sets are). It also edges the model's own self-ranking (14/19, 73%), but that +11-point margin is a two-compound difference at n=19 and is **not** statistically significant

(McNemar exact p≈0.63); the load-bearing claim is that verification precision is high in absolute terms while recall binds — not that forward-verification beats self-ranking. The overall top-1 moves only 23%→26% because the binding constraint is **generation recall**: the true structure was never proposed for 41 of 60 compounds, which no re-ranking can repair.

LLM elucidation therefore factorises into two near-independent levers: the **verifier is already strong (84%)**; the **generator (31% recall) is the wall** (Fig. 1).

5.3 Generate-wide: testing the recipe

The decomposition implies a recipe — *generate wide, verify by forward prediction* — a chemistry analog of self-consistency sampling, where many independent generations are pooled and the answer is chosen by agreement rather than from a single pass.[24] We tested it directly: ten independent solver agents proposed up to **six regiochemistry-aware candidates per compound**, pooled with the originals, and the 65 new candidates were forward-predicted and re-ranked as before. The measured result:

Table 6. Generate-wide vs original on the same 60 compounds.

	original	generate-wide
recall (true structure among candidates)	31%	**41%**
forward-verified top-1	26%	**30%**
verification precision (conditional on recall)	84%	72%

Wide generation lifts recall +10 points (31%→41%) and exact top-1 +7 points over the self-ranking baseline (23%→30%) — equivalently +4 over the original forward-verified top-1 (26%→30%, Table 6) — on the same 60 compounds (Fig. 5); a real, measured gain with no training. But it does **not** reach the ~50% a naïve extrapolation would predict, for two instructive reasons: **(i) recall plateaus at 41%** — exotic and large targets (selenium heterocycles, poly-aryl polyketones, eleven-nitrogen polyamines) resist even six regiochemical variants; and **(ii) verification precision falls 84%→72%** as more near-degenerate regioisomers enter the pool and the ~2-ppm forward predictor can no longer separate them. This recall/precision tension is the honest ceiling of the training-free approach: the recall-bound diagnosis stands, and closing the gap further requires either sharper verification or 2D-NMR constraints, not merely more candidates.

5.4 Does a deterministic verifier help? A HOSE-code ablation

§5.3 and a natural reading of the verifier-precision story suggest an obvious fix: swap the LLM forward-predictor for a *deterministic* ^{13}C predictor with a rigorous error model. We tested the canonical choice — a **HOSE-code[25]-style lookup trained on nmrshiftdb2[26]** (RDKit radial-environment bins, spheres r=4→1 with a hybridisation prior fallback; 31,000 molecules, 332,595 assigned carbons; held-out **MAE 3.23 ppm, median 1.73 ppm**) — as a drop-in replacement for the verifier, re-ranking the same §5.2 candidates.

It does **not** help. The HOSE verifier recovers **14/19 (73%)** of the in-set compounds — identical to the solver's own ranking and *below* the LLM verifier's **16/19 (84%)** on the same set. The reason is coverage, not the method: of the 2,244 candidate carbons, only **2%** match a training environment at the most specific sphere (r=4) and **67%** resolve only at coarse spheres (r≤2), because the benchmark's exotic chemistry (selenium heterocycles, polyaryl ketones, large polyamines) is under-represented in nmrshiftdb2. A coarse environment cannot separate the regioisomers the verifier exists to separate, so the deterministic predictor degrades to the self-rank baseline exactly where discrimination is needed — whereas the LLM forward-predictor generalises across novel scaffolds. The practical lesson is the opposite of the naïve one: on hard, real chemistry the LLM's *breadth* is an asset for verification. We show next (§5.7), however, that the failure is **method, not only coverage**: a *learned* ^{13}C model trained on the *same* nmrshiftdb2 data generalises across these environments and recovers the LLM verifier's precision — so a cheap learned predictor, not necessarily DFT-level shift accuracy or 2D-NMR, suffices to *match* the LLM verifier. What none of these reach is the near-degenerate-regioisomer precision ceiling (§5.3/§5.5), where DFT-level accuracy or orthogonal 2D-NMR constraints remain the genuine fix.

5.5 Negative control

Permutation negative control (Y-randomisation analog). A re-ranker earns the interpretation that it exploits genuine predicted-vs-observed ^{13}C agreement only if that agreement is destroyed when the pairing is broken. Borrowing the Y-randomisation discipline of QSAR validation, we re-paired — as a *derangement*, so no compound keeps its own spectrum — which observed ^{13}C spectrum each candidate set is scored against, and re-ran the verifier 1,000 times. Conditional-on-recall precision falls from the true **84.2% (16/19)** to a permuted mean of **66.4%** (95% range 52.6–78.9%; one-sided empirical p=0.019, two-sided p=0.038) — the verifier acts on real spectral agreement, not a candidate-list artefact. The honest caveat is the height of the chance floor: because recall-positive compounds carry few and near-identical (regioisomeric) candidates, even a random pairing lands on the correct structure ~66% of the time, so the verifier's genuine margin over chance is ~**18 points**, not the full 84% — real and significant, but to be read against this high baseline.

Confidence calibration (a negative result). We also asked whether the chamfer match-distance doubles as a confidence signal that would support selective prediction. It does not. Ranking the 48 multi-candidate compounds by their chamfer margin (the gap between the best and second-best predicted-vs-observed distance) and answering only the most-confident fraction leaves top-1 essentially flat and non-monotonic with coverage ($\approx$21% overall, neither rising nor falling reliably as the threshold tightens). Compounds with a single proposed candidate have no margin at all and must be excluded; an earlier analysis that retained them produced a spurious "improvement" that was entirely an artefact of those trivial cases. We therefore do **not** claim the verifier distance as a calibrated abstention gauge — a sharper, DFT/2D-NMR-grounded confidence estimate (§5.4) remains the route to reliable selective prediction.

5.6 Is the recall wall task-intrinsic? A trained-generator probe

The ceiling above is a property of *training-free LLM elicitation*; it leaves open whether the ~41% recall plateau is intrinsic to 1D-data elucidation or specific to how LLMs are elicited. We test this with a complementary probe — a small (~16M-parameter) ^{1}H/^{13}C$\rightarrow$SMILES transformer (simulated-spectra pretraining, then fine-tuned on IRexp; ensemble of four), held **deliberately separate from the training-free protocol**. Its candidates are filtered to the given molecular formula, pooled with Claude's, and re-ranked by the same verifiers.

The generator supplies candidates enumeration cannot. On the 194-compound benchmark the true structure enters the candidate pool for **54.1%** of compounds (versus 41.8% for scaffold enumeration and 33.5% for Claude alone). Crucially, where enumeration's near-degenerate regioisomers *collapse* the deterministic HOSE verifier (top-1 28.4%→16.0%, the §5.3 precision-loss mechanism), the generator's formula-correct, ^{13}C-separable candidates **convert**: HOSE top-1 rises **28.4%→35.1%** (McNemar exact p=0.015; +6.7 points, 95% CI [+2.1, +11.9]; Fig. S5). With the stronger LLM forward-prediction verifier on the 60 forward-verify compounds, recall rises 41%→**56%**; the corresponding forward-verified top-1 (measured at **41%**, precision 72%→73%) was obtained with blind in-house prediction agents and is reported as **provisional** pending deposit of those forward-prediction outputs and a re-run under the pinned model snapshot (the released `forward_verify_gen.py` reproduces the recall arm and the deterministic HOSE re-rank; the LLM forward-prediction JSONs are not yet in the bundle).

Two controls guard against memorisation, both reproducible from the released bundle (`contrib/generator_probe`): (i) the fine-tuning split removes every benchmark InChIKey-14 (train∩benchmark = 0, val∩benchmark = 0), and **none of the 40 newly-recovered compounds appear in either training stage** (0/40 in simulated pretraining, 0/40 in fine-tuning); (ii) the simulated-pretrained model alone recovers **0/248** benchmark structures zero-shot, rising to 25% only after IRexp fine-tuning. The IRexp data — not the architecture — is the active ingredient, and the recall ceiling is therefore **elicitation-specific, not task-intrinsic**. We report this as a probe, not part of the headline training-free protocol; it answers whether the wall is breakable, not whether to abandon training-free methods. (The released public split `irexp_release/train` does *not* hold out the benchmark — it overlaps it by 117/200 InChIKey-14 — so downstream users must de-leak as in `build_exp_manifest.py`; see Data and code availability.)

5.7 A learned ^{13}C verifier recovers the precision the lookup could not

§5.4 leaves one test unrun: the HOSE predictor is a *lookup*; would a *learned* model trained on the **same nmrshiftdb2 data** generalise across the under-represented environments where the lookup degrades? We train a small message-passing GNN (4 layers, per-carbon ^{13}C regression) on the identical dump that builds the §5.4 table; on a held-out split it predicts ^{13}C at **MAE 1.70 ppm** (median 1.02) — roughly twice as sharp as the lookup's 3.23 ppm, an independently-competent predictor. Dropped into the verifier slot on the **same §5.2 candidate set**, holding training data and evaluation fixed so that only the method changes:

Table 7. Verifier comparison, conditional on recall (n=19), on the identical §5.2 set.

verifier	top-1 | recall	held-out ^{13}C MAE
solver self-ranking	14/19 (73%)	—
deterministic HOSE *lookup* (§5.4)	14/19 (73%)	3.23 ppm
learned GNN (same data)	**16/19 (84%)**	**1.70 ppm**
LLM forward-verifier (§5.2)	16/19 (84%)	—

The learned model recovers the full **73%→84%** precision the lookup could not, matching the LLM verifier (Fig. S6). The §5.4 reading is therefore too strong: the deterministic verifier's failure was substantially **method** — generalisation across novel environments — not coverage alone, and a cheap learned predictor on the same data suffices to match the LLM, without the compound-specific DFT accuracy or 2D-NMR §5.4 reaches for. The GNN and the LLM reach 84% on *partly different* compounds (two independent verifiers concurring, not one), and the gain is **generalisation, not memorisation**: candidate–training overlap is **0/126** by InChIKey-14 (we confirm zero overlap against the entire nmrshiftdb2 database; every true structure and all three GNN-wins are absent), Morgan-Tanimoto nearest-neighbour to training is a modest 0.44 median with the wins sitting *at* — not above — the median, and a Y-randomisation control (1,000 derangements) places the real 84% above the 97.5th percentile of the chance distribution (mean 58.6%, 95% range 42.1–73.7%; one-sided $p<0.05$).

We report this as a trained complement, like §5.6. The decisive comparison is n=19, so the +2 over the lookup is **confirmatory of the LLM number, not a precise quantitative gain** (one decisive win at 0.79 ppm, two tight, one 0.08-ppm coin-flip loss). Near-degenerate regioisomers (e.g. R22/R26/R28) sit at or below the ~1–2 ppm forward accuracy and are resolved only by coin-flip margins, so the precision ceiling of §5.3/§5.5 stands — the GNN reaches the LLM verifier's quality *cheaply*, it does not break that ceiling. Artifacts and both leakage checks are released (`scripts/gnn_predict.py`, `data/fverify/gnn_results.txt`, `data/nmrshiftdb/gnn_c13.pt`; `docs/VERIFIER_PROBE.md`).

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6. Discussion

The frontier LLMs we tested (the Claude family) are neither "solving structure elucidation" on realistic 1D data nor failing at it. They are reliable **scaffold-level** elucidators (best-candidate Tanimoto ≥ 0.45 for 56% of compounds, 73% of simple ones; mean best Tanimoto 0.59) and good **verifiers** (84% conditional on recall). Exact recovery is throttled by candidate recall and by the regiochemical underdetermination intrinsic to 1D NMR. The contribution of this paper is therefore best read as a **diagnosis with a bounded, training-free improvement attached**, not as a method that solves the task. The diagnostic result is the durable take-away: across models, domains, and verifiers, the wall is **generation recall**, not verification. The accompanying improvement is real but deliberately modest — forward verification with wide generation lifts top-1 from 23% to 30%, and we show by direct measurement (§5.3) that this stays *below its own ceiling* (recall plateaus at 41%; verification precision falls to 72% as near-degenerate regioisomers accumulate), while the match distance fails as a confidence gauge for selective prediction (§5.5, a reported null result). This reframes the engineering problem. Rather than training a bespoke spectra→structure model — a target the frontier already meets at the scaffold level and that ages out each model generation — the durable levers are (i) **open, hard, honestly scored benchmarks** that expose specific failure modes (here, regiochemistry and recall), and (ii) **inference-time scaffolding** (decoupled solving, generate-and-verify) that needs no training and improves with each model. IRexp's IR modality and 2D-NMR-ready provenance position it for the obvious next step: supplying the HMBC/COSY/NOESY constraints that would attack regiochemistry at the source. Our trained-generator probe (§5.6) sharpens the point: the recall wall is *elicitation-specific, not task-intrinsic* — a small generator fine-tuned on IRexp lifts it (recall 41→56%) — and because that gain comes entirely from the released data (0→25% with fine-tuning, 0/248 without), it is the **open dataset**, not a bespoke architecture, that does the work when training does help.

Protocol over fashionable levers. That *how* a problem is posed and verified — decoupled per-compound solving, generate-and-verify — moves accuracy more than raw model capability does echoes a recurring lesson across cheminformatics: careful pipeline and protocol design tends to dominate the fashionable component. The same pattern appears in molecular property prediction, where a recent systematic benchmark reports that learned, pretrained molecular embeddings rarely outperform classical ECFP fingerprints once evaluation is controlled[27] — the modelling fashion underperforms the well-engineered baseline. We read our within-compound control (§4.3) and forward-verification recipe (§5) in the same light: the durable gains in LLM elucidation come from honest benchmarking and inference-time scaffolding rather than from waiting on the next, larger model. We draw this parallel narrowly and claim no formal equivalence between the two settings.

That the bottleneck reproduces inside a single application domain (§4.5: 26% top-1 on battery-electrolyte chemistry, n=46, against 28% overall) is consistent with it being structural rather than incidental — and it makes the forward-verification recipe directly relevant to the magnetic-resonance workflows (electrolyte-decomposition assignment, NMR crystallography) where computing a candidate's spectrum to confirm it is already standard practice.

For practitioners, two operational findings transfer immediately: solve each problem in a fresh context with tool access (5%→15%, ≈3×, §4.3), and trust a proposed structure in proportion to how well its forward-predicted spectrum matches the observed one — while remembering the abstention caveat below.

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7. Limitations

Several limitations of earlier drafts are now resolved with controlled experiments; we state what remains plainly.

Resolved. **Extraction noise:** an RDKit self-consistency audit (^{13}C peak count vs symmetry-unique carbons, formula, SELFIES round-trip) finds **57/60 ground truths spectrally clean**, and the metrics are unchanged on that clean subset (self-ranking top-1 24% vs 23%, recall 33% vs 31% on the n=60 forward-verify set) — the conclusions are not driven by scraping artefacts. **Verifier precision / abstention:** the generate-wide experiment (§5.3) quantifies the recall/precision tension directly — verification precision is 72–84% conditional on recall and degrades as near-degenerate regioisomers accumulate, so forward-match distance is a strong *re-ranker* but a soft *confidence* gauge. We further tested non-LLM verifiers (§5.4, §5.7): a nmrshiftdb2-trained HOSE-code *lookup* does **not** beat the LLM verifier (73% vs 84% conditional), but a *learned* GNN on the same data **does** recover it (84%, §5.7) — so the deterministic failure was method, not only coverage, and a cheap learned predictor matches the LLM verifier without DFT-level shifts or 2D-NMR. What remains beyond all of them is the near-degenerate-regiochemistry precision ceiling, where DFT-level accuracy or 2D-NMR constraints are still the genuine fix. **Projection:** §5.2's extrapolation is replaced by the **measured** §5.3 result (top-1 30%, recall 41%) — and is honestly below the optimistic estimate.

Independence checks. Scoring throughout is mechanical RDKit (not LLM-judged), and all solver/verifier runs were transcript-audited *at generation time* for zero web and zero ground-truth access; the committed artefacts are the parsed per-compound predictions, and the underlying transcripts are available on request. To confirm that the forward-verifier's measured advantage reflects real predicted-vs-observed spectral agreement rather than leakage or a candidate-list artefact, we ran a permutation negative control (Y-randomisation analog, §5.5): re-pairing — as a derangement — which observed ^{13}C spectrum each candidate set is scored against, over 1,000 permutations, collapses conditional-on-recall precision from 84.2% to a chance mean of 66.4% (two-sided p=0.038) — the signal weakens to the chance floor under label-shuffle, as a correctly isolated blind evaluation requires. A second Claude model (Sonnet) re-solved a 12-compound subset under the identical generate-wide protocol and was comparably recall-bound (recall 33% vs Opus 41% on those compounds), consistent with the recall bound being a property of the task rather than of one model. As this is within the Claude family, it speaks to model-instance robustness, not cross-vendor generality.

Remaining. **(i) Human audit — prepared but not yet run.** Solver and verifier are both LLMs, so the one validation we cannot perform ourselves is an expert-chemist review of a sample of elucidations and forward predictions. We have therefore **built and frozen** a blinded, pre-registered audit package — a difficulty-stratified 30-compound sample (9 recall-positive), rendered candidate structures, a separate withheld answer key, and a mechanical scoring sheet — released at `data/audit/` with its protocol at `docs/EXPERT_AUDIT_PROTOCOL.md` (regenerable via `scripts/make_audit_sample.py`), designed to test the two load-bearing claims (misses are predominantly consistent regioisomers, §4; forward verification is a trustworthy re-ranker, §5). **We have not yet run the panel**; until expert results are in, those claims should be read as machine-validated (RDKit InChIKey) but not yet human-validated.

(ii) Single vendor and an underpowered cross-model comparison. Every number comes from one vendor's models. The headline (28.4% top-1) is a single frontier model (Claude Opus), and the cross-model evidence (§4.4) is four **Claude-family** models on a shared **24-compound** subset. That comparison is statistically underpowered: only the extreme Fable-vs-Haiku contrast survives multiple-comparison correction (Holm-adjusted p=0.006); Fable-vs-Opus is not significant (uncorrected p=0.063, Holm-adjusted p=0.19) and Opus-vs-Sonnet are indistinguishable, with overlapping bootstrap CIs. What is robust is the **ranking** — outcomes are strictly nested (Haiku $\subset$ Sonnet $\subset$ Opus $\subset$ Fable) and the weakest floors at 0% — so the benchmark is capability-sensitive; what is *not* established at n=24 is the size of any adjacent gap. Quantitative claims should therefore be read as holding **for the Claude family**, and a true cross-**vendor** sweep (GPT-, Gemini-class, open-weight models) — the test of whether the pattern is a property of the task rather than one lineage — was forgone because it needs paid API access incompatible with our zero-cost protocol; we flag it as the most important external-validity experiment left open.

(iii) Domain-subset scope: the battery-electrolyte case study (§4.5) comprises *literature compounds bearing electrolyte-relevant functional chemistry* drawn from the open corpus — not operando or in-cell decomposition spectra — so it demonstrates functional-class transfer of the elucidation bottleneck, not direct assignment of authentic interphase/degradation products, which would require a dedicated operando set.

(iv) Constitution-only scoring: correctness is judged on InChIKey connectivity, so a correct-constitution / wrong-stereochemistry prediction is counted correct. This affects at most the 10.3% (20/194) of targets with a defined stereocentre and is intrinsic to what 1D NMR/IR can determine; a stereochemistry-sensitive benchmark would need 2D/chiroptical data.

(v) Single-sample scoring: each headline compound is scored from one solver prediction set (one decoupled run), so reported top-1/recall carry no run-to-run (LLM-sampling) variance estimate — the bootstrap CIs reflect compound sampling only. The generate-wide experiment (§5.3) pools ten independent generation passes (recall 31%→41%), indicating generator stochasticity that single-pass scoring understates.

(vi) Chemical-space coverage: the benchmark is drawn from open-access organic methodology and total-synthesis literature, covering small-to-medium drug-like and synthetic-organic space; it is **not** representative of organometallic/coordination compounds, large biomolecules (peptides, oligonucleotides, oligosaccharides), or stereochemistry-heavy targets, and our results should not be extrapolated to them.

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8. Methods

Mining and resolution. PMC-OA full text was fetched from `s3://pmc-oa-opendata`; records were extracted with a deterministic parser and resolved with OPSIN, RDKit, and SELFIES, with an optional cached PubChem fallback. Durable content-keyed caches make resolution additive and restart-safe.

Benchmark and agents. Problems were sampled from `irexp_resolved`, stratified by RDKit ring analysis, with cross-round de-duplication by InChIKey. The battery-electrolyte subset (§4.5) was drawn from the same corpus by SMARTS substructure filters for six electrolyte functional classes (carbonate, sulfonyl/sulfonate, nitrile, sp^3-C–F, phosphoryl, glyme/oligoether), balanced to eight compounds per class (48 curated; 46 scored after two yielded no parseable candidate), excluding every compound used elsewhere, and J-enriched and spectrally validated identically to the main rounds. Solver and forward-prediction agents were independent Claude-Opus sub-agents invoked under a consumer subscription; agents were instructed closed-book and audited via automated transcript search for zero web/answer access. Scoring used RDKit InChIKey-connectivity matching and Morgan(2, 2048) Tanimoto; forward-verification used a symmetric chamfer distance over ^{13}C peak sets. The core protocol trains no model and uses no paid API; the two trained probes — the §5.6 generator and the §5.7 learned ^{13}C verifier — are the only separately-trained exceptions, each reported as a complement and fenced from the headline results.

Models and versions. All experiments used Anthropic Claude models via the consumer subscription (claude.ai). The §4.4 comparison spanned, in capability order, Claude Haiku, Claude Sonnet, Claude Opus, and Claude Fable 5; the headline benchmark (§4.1) and forward-verification (§5) used Claude Opus. Exact model snapshot identifiers and the access window are recorded in `docs/MODELS.md` so the benchmark can be re-run against the same checkpoints.

Reproducibility. Every round is frozen: questions, ground-truth answers, per-agent raw outputs, predictions, and scorer outputs are released, and the sampler, scorer, and forward-verification harness are scripted end-to-end.

Figures (main text)

- **Fig. 1** (`docs/figures/fig_wall.png`) — the diagnosis in one glance: shown a candidate set the model *verifies* the true structure 84% of the time, yet *proposes* it for only 31% of compounds. Recall, not verification, is the wall (53-point gap).
- **Fig. 2** (`docs/figures/fig1_difficulty.png`) — top-1 and recovered accuracy on IRSpectra-Bench by difficulty (all / simple / complex, n=194) with bootstrap 95% CIs.
- **Fig. 3** (`docs/figures/fig5_models.png`) — four-model comparison on a 24-compound subset: Fable 5 45% > Opus 25% > Sonnet 20% > Haiku 0% top-1 (strictly nested); capability-sensitive but underpowered to separate adjacent models at n=24.
- **Fig. 4** (`docs/figures/fig_mechanism.png`) — forward-verification on a real benchmark regioisomer pair (picolinamide vs nicotinamide): forward-predicted ^{13}C matches the true isomer (chamfer 0.42 vs 1.30 ppm), an analog of NMR-crystallography.
- **Fig. 5** (`docs/figures/fig3_method.png`) — forward-verification inference ladder on the same 60 compounds: solver self-ranking → + forward-verify → + generate-wide

(23%/26%/30% top-1).

Supporting Information figures

- **Fig. S1** (docs/figures/fig0_overview.png) — study design: open multimodal data (IRexp) → blind, complexity-stratified benchmark → decoupled blind solving → forward-verification re-ranking; training-free core pipeline.
- **Fig. S2** (docs/figures/fig4_dataset.png) — IRexp composition: IR records → NMR-paired → structure-linked → full IR+^{1}H+^{13}C+structure quadruples.
- **Fig. S3** (docs/figures/fig2_size.png) — accuracy vs molecular size (heavy-atom bucket); the monotonic 60%→7% top-1 gradient.
- **Fig. S4** (docs/figures/fig6_electrolyte.png) — top-1 and recovered accuracy on IRSpectra-Bench-Electrolyte by battery-electrolyte chemical class (n=46): sp^{3}-C–F easiest (50%), sulfonyl and nitrile hardest (12%); overall 26%/28%.
- **Fig. S5** (docs/figures/fig_generator_probe.png) — trained-generator probe (§5.6; a complement, not part of the training-free protocol): candidate recall and deterministic-HOSE top-1 on the 194-compound benchmark for Claude / + scaffold enumeration / + trained generator. Enumeration's near-degenerate isomers collapse the verifier (28.4→16.0%) while the generator's formula-correct candidates convert (28.4→35.1%).
- **Fig. S6** (docs/figures/fig_verifier.png) — learned-verifier probe (§5.7; a complement, not part of the training-free protocol). (A) Conditional-on-recall top-1 (n=19) for the four verifiers: a GNN trained on the same nmrshiftdb2 data as the HOSE lookup recovers the LLM verifier's 84% that the lookup (73%) could not. (B) Why: held-out ^{13}C MAE — the learned model is ~2× sharper (1.70 vs 3.23 ppm).

Data and code availability

All data and code are released in the project repository: IRexp and the irexp_resolved split (data/irexp/, data/irexp_resolved/); the benchmark rounds and within-compound control (data/benchmark*/, scored by scripts/benchmark_v2.py); the battery-electrolyte case-study subset (data/benchmark_electrolyte/, built by scripts/build_electrolyte_bench.py, scored by scripts/score_electrolyte.py); the ground-truth integrity audit (scripts/validate_benchmark.py, data/benchmark*/clean_qids.json); the forward-verification and generate-wide experiments (data/fverify/, data/gw/, data/fverify2/, scripts/forward_verify.py); the deterministic HOSE-code verifier ablation (scripts/hose_predict.py, data/fverify/hose_results.txt); and the dataset-mining pipeline (spectro_scraper/). Companion technical notes: docs/BENCHMARK.md and docs/FORWARD_VERIFY.md. The verifier negative-control and selective-prediction analyses (scripts/verifier_diagnostics.py), the recall-headroom and scaffold-enumeration study (scripts/analyze_recall_headroom.py, scripts/enumerate_isomers.py, scripts/closing_the_gap.py), the modality-ablation harness (scripts/modality_ablation.py, docs/MODALITY_ABLATION.md), and the blinded expert-audit package (data/audit/, docs/EXPERT_AUDIT_PROTOCOL.md) are released likewise. The §5.6 trained-generator probe ships as a self-contained bundle (contrib/generator_probe/): generator candidates, the de-leaked split with its InChIKey-14 manifest, and the verification scripts (scripts/closing_the_gap_gen.py, scripts/forward_verify_gen.py, scripts/verify_leakage_exact40.py); model checkpoints are deposited on Zenodo. Downstream note: the public irexp_release/train split does **not** hold out IRSpectra-Bench (117/200 InChIKey-14 overlap); de-leak with contrib/generator_probe/build_exp_manifest.py before training models that will be evaluated on the benchmark. The §5.7 learned-verifier probe ships its full reproducer — `scripts/gnn_predict.py` (extract/train/score/control), the trained model `data/nmrshiftdb/gnn_c13.pt`, per-compound results and both leakage checks (`data/fverify/gnn_results.txt`), and the write-up `docs/VERIFIER_PROBE.md`; the GNN trains on the same nmrshiftdb2 dump as the §5.4 HOSE lookup. **Archival deposit:** a complete frozen snapshot (dataset, benchmark, answer keys, predictions, scripts, figure-regeneration, and the expert-audit package) will be archived on Zenodo under DOI [**TODO: 10.5281/zenodo.XXXXXXX — mint on submission**]; the GitHub repository is the development mirror and the Zenodo record the citable version of record. (The held-out answer keys `data/audit/key.jsonl` and `data/modality/key.json` are deposited but flagged "withhold from blinded reviewers".)

Licensing and attribution

IRexp is derived from two open-access source pools and redistributed under terms compatible with each (see `data/NOTICE`). **(a) Redistribution:** we release only *extracted numeric data* — IR band lists, $^{1}H/^{13}C$ shift lists, and resolved structures (SMILES/SELFIES/InChIKey) — plus each record's source DOI/accession, not source full text, figures, or PDFs. **(b) Two separable pools:** 119,345 records derive from the PMC Open-Access Subset (**CC-BY-4.0**) and 1,888 from the Chemotion RADAR4Chem FT-IR deposit (**CC-BY-SA-4.0**); the pools are kept separate and individually labelled, so users may take the CC-BY pool alone, while any combined or Chemotion-derived release carries CC-BY-SA-4.0 to honour the ShareAlike term. Code is released under the MIT License. **(c) Attribution:** re-users must cite this dataset (Zenodo DOI above) and attribute the original publications via each record's `source_doi`.

Author Contributions

I.Y.: conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing — original draft. **R.S.:** methodology, software, formal analysis, investigation, validation (trained-generator and learned-verifier probes, §5.6–§5.7), writing — review and editing. **R.A.V.-H.:** conceptualization, methodology, supervision, writing — review and editing.

Conflicts of Interest

The authors declare no competing interests.

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Figure plates

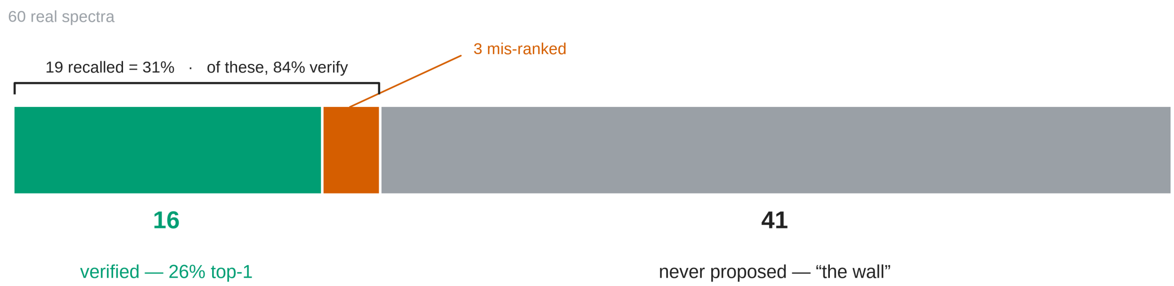

Fig. 1: The diagnosis as a single part-to-whole bar of the 60 forward-verify compounds. The true structure is verified top-1 for 16 (green), recalled but mis-ranked for 3 (vermilion), and never proposed for 41 (grey) — *the wall*, 68% of the bar. Forward-verification recovers 16/60 = 26% exact top-1 end-to-end: the model proposes the true structure for only 19/60 = 31% of compounds and, of those, verifies 84%. Recall, not verification, is the wall.

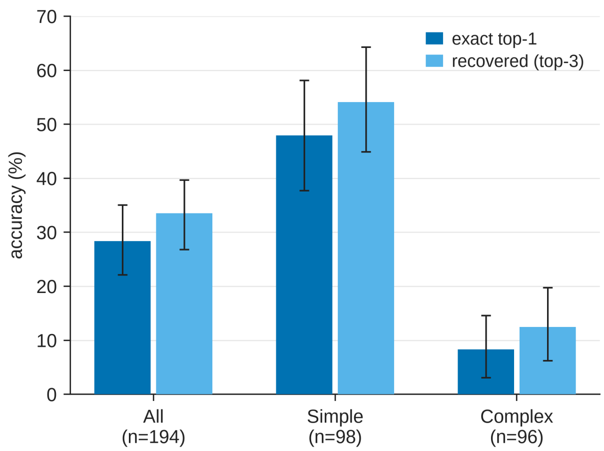

Fig. 2: Top-1 and recovered accuracy on IRSpectra-Bench by difficulty (all / simple / complex, n=194) with bootstrap 95% CIs.

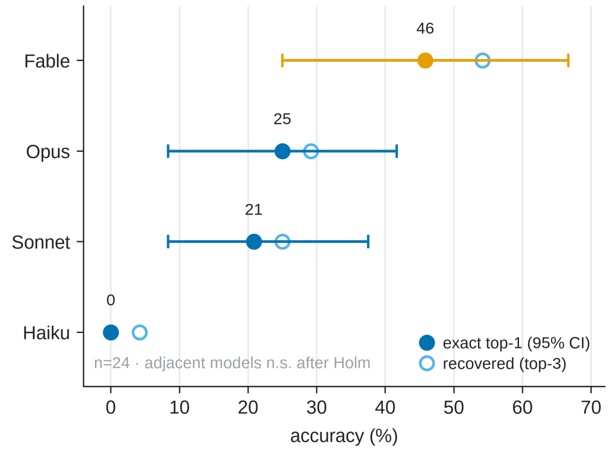

Fig. 3: Four-model comparison on a 24-compound subset: Fable 5 45% > Opus 25% > Sonnet 20% > Haiku 0% top-1 (strictly nested; underpowered to separate adjacent models at n=24).

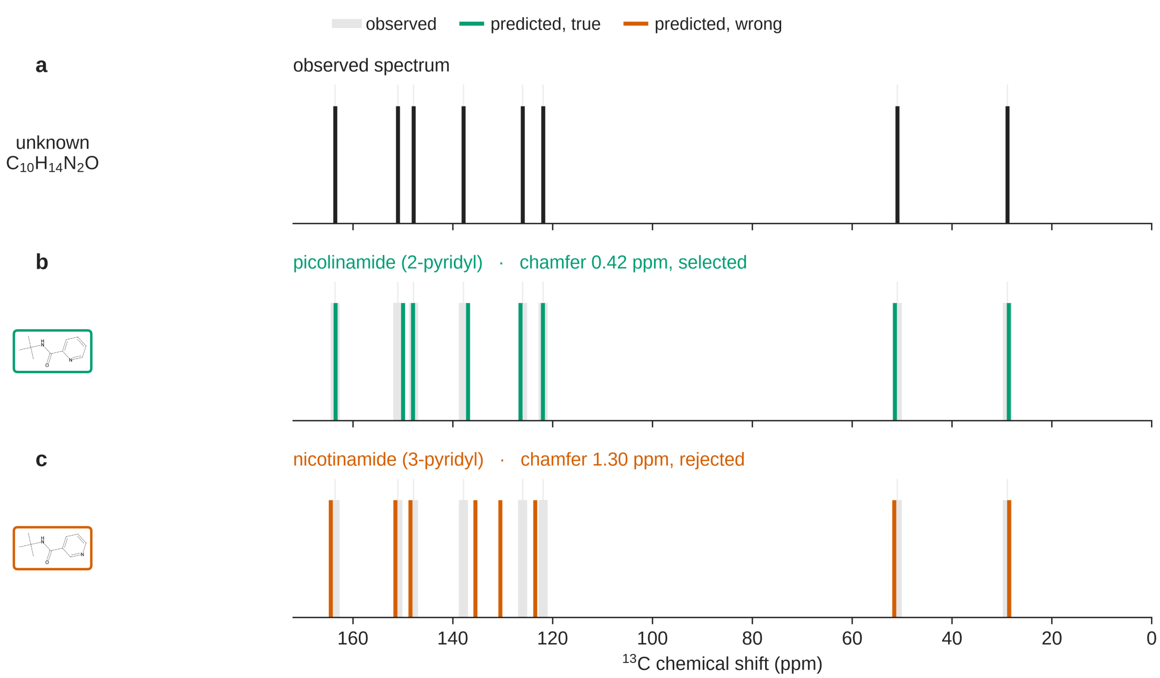

Fig. 4: Forward-verification on a real benchmark regioisomer pair (picolinamide vs nicotinamide): forward-predicted ^{13}C matches the true isomer (chamfer 0.42 vs 1.30 ppm) — an analog of NMR-crystallography.

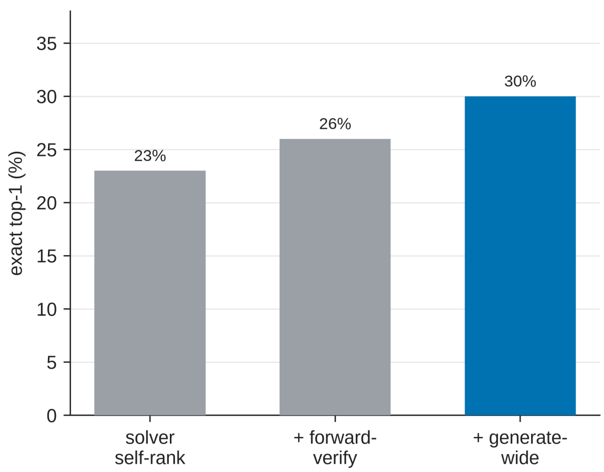

Fig. 5: Forward-verification inference ladder on the same 60 compounds: solver self-ranking → + forward-verify → + generate-wide (23%/26%/30% top-1).

Supporting Information figures

Fig. S1: Study design: open multimodal data (IRexp) → blind, complexity-stratified benchmark → decoupled blind solving → forward-verification re-ranking; training-free core pipeline.

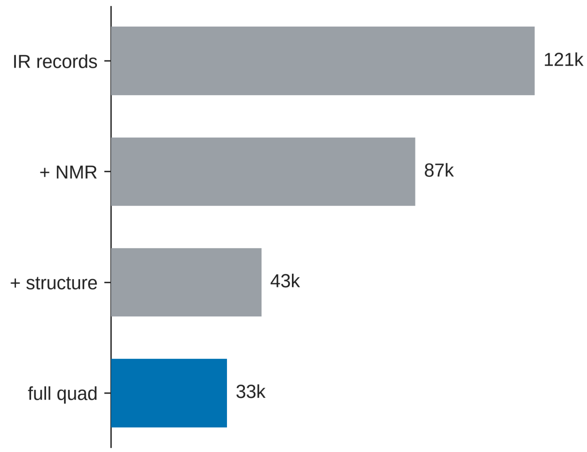

Fig. S2: IRexp composition: IR records, NMR-paired, structure-linked, and full IR+^{1}H+^{13}C+structure quadruples.

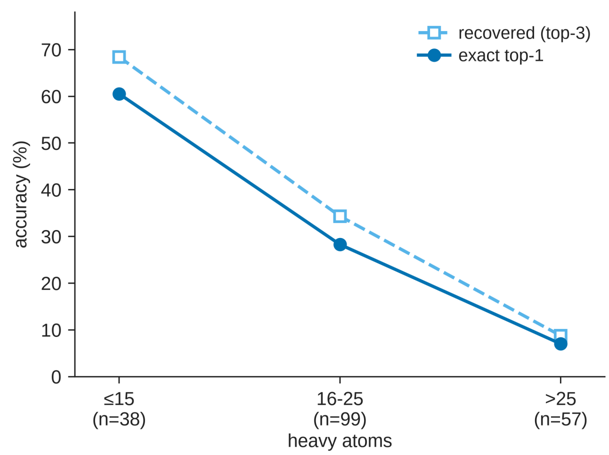

Fig. S3: Accuracy versus molecular size; the monotonic 60%→7% top-1 gradient with heavy-atom count.

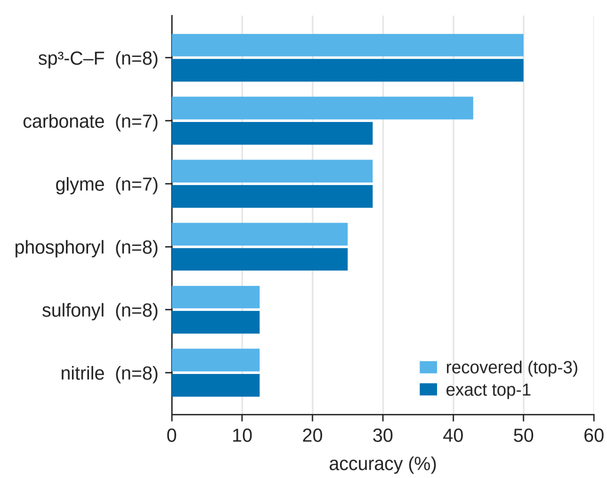

Fig. S4: IRSpectra-Bench-Electrolyte by battery-electrolyte class (n=46): sp^3-C–F easiest (50%), sulfonyl and nitrile hardest (12%).

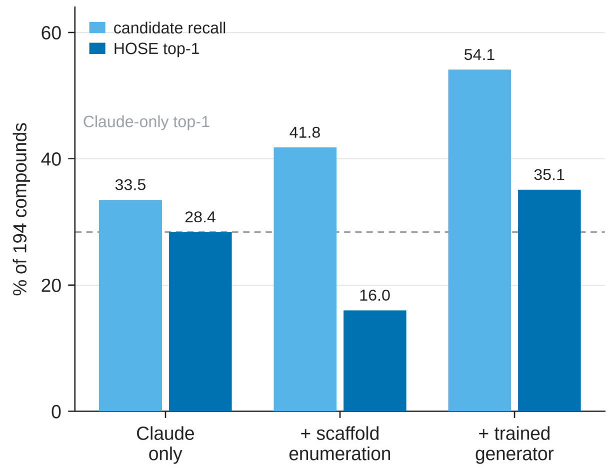

Fig. S5: Trained-generator probe (§5.6; a complement, not part of the training-free protocol). Candidate recall and deterministic-HOSE top-1 on the 194-compound benchmark for Claude / + scaffold enumeration / + trained generator: enumeration's near-degenerate isomers collapse the verifier (28.4→16.0%) while the generator's formula-correct candidates convert (28.4→35.1%).

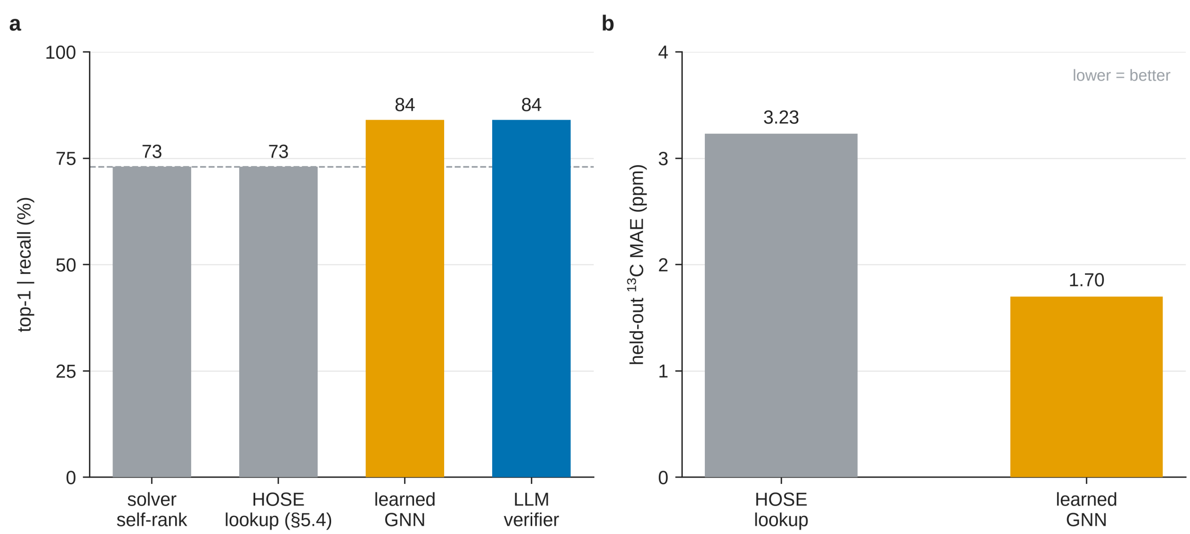

Fig. S6: Learned-verifier probe (§5.7; a complement, not part of the training-free protocol). (A) Conditional-on-recall top-1 (n=19) across four verifiers: a GNN trained on the same nmrshiftdb2 data as the HOSE lookup recovers the LLM verifier's 84% that the lookup (73%) could not. (B) Held-out ^{13}C MAE — the learned model is roughly 2× sharper (1.70 vs 3.23 ppm).